

ChartAuth

Eligibility verification and the evidence behind completion

A healthcare task can receive full credit without achieving its purpose. I found that gap in a clinical-trial evaluator, then built an eligibility workflow that requires the patient, source response and saved result to agree. The design rewards a supported next action, including a documented handoff when coverage cannot be resolved.

1 Understand the environment

I selected HealthAgentBench `clinical_trial_matching_task_19`: a document task with one patient, 301 trials and four eligible IDs. Its explicit output makes the reward auditable. I inspected the task, environment and evaluator at commit `bcbb8085`. Six controlled probes exercise the evaluator directly; I did not run the full harness.

Element	What the inspected task does
Agent observations	Patient note, topic and trial XML staged in the workspace.
Actions and state	Terminal commands read documents, run scripts and write the submission. Files and processes change; no care transaction is modeled. The data mount is writable.
Hidden information	Gold labels and grading code are intended to remain outside the agent service. Effective isolation and access to public answers still need a sandbox test.
Initialization and termination	Bootstrap stages inputs. Configuration allows 3,600 seconds for the agent and 300 for grading. The output is a confidence-ordered list of trial IDs; runtime lifecycle behavior was not tested.
Success rule	Deduplicate IDs and discard those outside the candidate pool. Award 1 if all four eligible IDs occur in the first 50 retained predictions; otherwise 0. Precision is reported but does not gate reward.

Both a benchmark and an RL environment

The benchmark already has observations, actions, state transitions and an episodic reward. Reinforcement learning does not require dense rewards or a built-in trainer. To train responsibly, I would verify clean resets and termination, protect labels and grading, repair the reward, and connect a rollout loop. Fixed held-out tasks and a reporting protocol preserve its benchmark role.

What the contrasting designs contribute

MedAgentBench offers structured EHR interactions. In its inspected POST branch, JSON is parsed and success is reported without an HTTP POST in that branch. I would verify persistence before trusting that reply; separately supplied reference checks were not inspected. HealthAdminBench combines browser tasks with exact checks and judgment rubrics. I borrow that separation of transaction facts from interpretation.

Terms An episode is one bounded attempt. A benchmark compares results. A reward is the score used to reinforce behavior.

Sources [Task](#) / [Evaluator](#) / [Mounts](#) / [Configuration](#) / [Bootstrap](#) / [MedAgentBench](#) / [HealthAdminBench](#)

Audit the training signal

2 What the score accepts

The instruction asks for eligible trials and no others. Yet four correct choices plus 46 incorrect choices earn full credit: $4 / 50 = 8\%$ precision. The reward is binary, so there is no partial-credit tier to audit. These probes test scoring behavior, not an agent's clinical performance.

Shortcut	Why it earns credit	Fix and new failure mode
Over-select trials Demonstrated	All four gold IDs fall within the first 50.	Reward: enforce the intended set or a reviewed precision constraint. Incomplete labels could reject valid trials.
Negate the correct IDs Demonstrated	The parser extracts an ID even from a sentence saying the patient is ineligible.	Deterministic parser: require the documented line format. Benign formatting may then fail.
Add unknown IDs Demonstrated	Out-of-pool IDs disappear before scoring.	Validation: reject unknown IDs and version the pool. A stale pool could reject a new valid trial.
Omit supporting evidence Accepted by construction	Only IDs are required; correct answers need no criterion-level support.	State and rubric: bind each criterion to source evidence. Superficially relevant citations can still mislead.
Retrieve public labels Unproven access path	Internet is enabled and labels are public; reachability was not tested.	Permissions: isolate grading and restrict retrieval; keep held-out cases private. Restrictions may block legitimate research.

Six reproducible checks

Controlled input	Reward	Retained precision
Four gold IDs only	1	100.0%
Gold plus 46 incorrect IDs	1	8.0%
All 301 IDs with gold first / gold last	1 / 0	1.33% in both
Negated gold-ID statements	1	100.0%
Gold plus 100 out-of-pool IDs	1	100.0% after filtering

The probes use synthetic IDs with the unchanged pinned evaluator. The two 301-ID orderings are separate checks; only their order changes. Neither the unused precision constant nor its stale docstring controls the executed reward.

Repair the narrow defect first

Fix parsing and the acceptance rule before adding a broad judge. Then validate labels and isolation. A deterministic verifier can still enforce a wrong reference, a public case can be memorized, and a source link does not prove support. Independent review, private case families and adversarial examples remain necessary before training.

Sources [Task instructions](#) / [Executed scoring logic](#) / [Environment permissions](#)

Produce reliable eligibility episodes

3 The environment and production plan

I chose a mock ModMed-style practice because specialty visits, insurance and staff handoffs belong in one workspace. The aim is a reliable coverage picture before care, supporting access and coordination. Better outcomes or value-based-care savings are not established by this sample.

Task and completion

Reset shows the patient, coverage card, visit and policy. The reference stays server-owned; query and poll reveal the payer response. Actions are inspect, query, poll, assess, save and finish. Finish ends the episode; action requests enforce a 40-action and 10-minute budget. Done means one saved assessment with the supported next action and owner. Active coverage never guarantees payment.

Three starter cases change one decisive fact: A records a \$30 copay and required authorization, then routes to authorization; B changes coverage to inactive and requests updated insurance; C changes the returned member and routes identity review. B and C leave unusable benefits unknown.

What is checked and who decides

Fifteen deterministic checks cover identity, response binding, 14 assessment fields, five current source references, exactly one saved record and guarded action order. All must pass for reward 1. Document digests detect changes, not truth. A future narrative rubric scores support and handoff usefulness from 0 to 2. A specialist resolves ambiguous records and policy meaning; a calibrated model judge could triage narrative but never override a hard failure.

Source lineage and quality gates

An eligibility specialist freezes sources and maps each reference fact to evidence. Two reviewers label independently. An engineer checks schema, resets and critical mutations. Release requires authorized sources, privacy review, coherent dates, a passing reference and failing unsafe variants. Measure pre-adjudication agreement, confusion matrices and Cohen's kappa. A proposed 50-case pilot targets 95% agreement and $\kappa \geq 0.80$; inspect prevalence and undefined kappa. A senior specialist adjudicates disagreements with a recorded rationale; unresolved cases stay quarantined.

Real charts require authorized use and documented Safe Harbor or Expert Determination de-identification. Review text, attachments and identifiers, isolate linkage keys, then revalidate chronology and labels. Date shifting alone is insufficient. Production evidence needs field-level source anchors; the sample binds whole documents and uses fictional records only.

Economics and expansion

Planning assumption	Initial	After 500
Labor minutes per candidate / accepted yield	125 / 85%	60 / 95%
Labor + compute per candidate	\$131.25 + \$5	\$65.00 + \$3
Hours / variable cost per accepted episode	2.45 / \$160.29	1.05 / \$71.58

Rates per hour: author \$45, each reviewer \$75, adjudicator \$100, engineering \$80. Both reviewers remain. Cost is (labor + compute) / yield. A separate \$6,000 setup allocation adds \$12 over 500 accepted episodes. These are planning point estimates, not a measured learning curve.

The sample has 50 templates: ten conditions across five services, with 1,000 generated visits. Its 84 tests include every visit and rejection of payment guarantees across all templates. Moving from 7 to 200 workflows across four EHRs creates up to 800 combinations before payer/specialty variation. Verifier maintenance, policy drift and reviewer calibration likely break first. Next: independent labels, private cases, ambiguous/no-response cases, adapter contracts and the 50-case pilot. Live EHR integration and training remain future work.

Sources [ModMed workflow](#) / [HHS de-identification](#)